K_L^0 identification efficiency of the KLM detector in the Belle II experiment using $_sPlot$

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Abstract

Long-lived neutral kaon mesons (K_L^0) are identified by energy deposits in the KLM detector of the Belle II experiment. However, various other particles can also deposit energy in the KLM detector. A statistical method known as $_sPlot$ can be utilized to distinguish between the energy deposit patterns of K_L^0 vs other backgrounds. This technique, currently developed with Monte-Carlo events, allows for a more robust determination of K_L^0 identification efficiency in the KLM detector, which can eventually be applied on real data.

Introduction

The KLM detector is the largest and outermost sub-detector in the Belle II experiment at the world's highest luminosity collider, the SuperKEKB, located in Tsukuba, Japan [1, 2]. The KLM, designed to identify long-lived neutral kaon mesons (K_L^0) and muons, consists of the Barrel KLM (BKLM) and Endcap KLM (EKLM). The BKLM shares its axis with the beam axis and has 15 layers, while the EKLM is placed at the two ends of the detector, normal to the beam axis [3, 4]. The detectors are made up of alternating layers of iron slabs and active elements. Whenever a K_L^0 moves through the detector, it deposits energy and is recorded as a cluster of hits in the KLM. Energy deposits in the KLM not associated with a track in the Central Drift Chamber are K_L^0 candidates as they are neutral particles, while those associated with a track are muon candidates since they are charged particles. The magnitude of the reconstructed K_L^0 energy is proportional to the size of a neutral KLM cluster, which however is a very rough approximation.

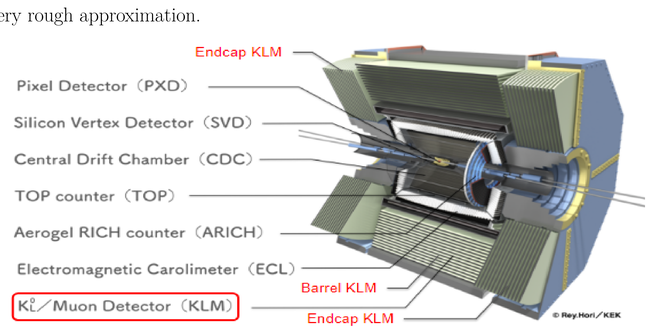


Figure 1: The Belle II detector.

Signal and tag decays of the control sample

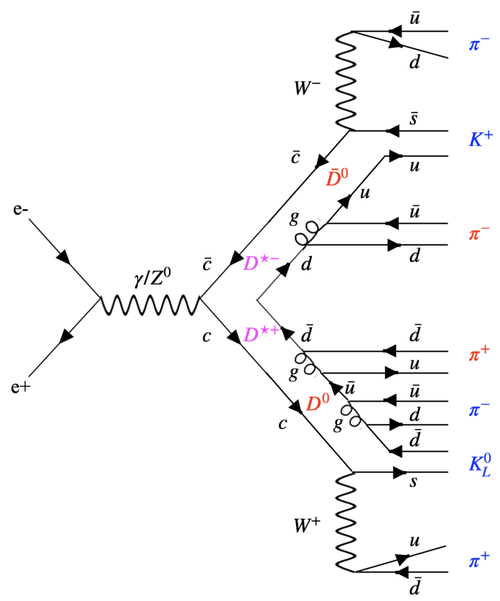


Figure 2: Feynman diagram of underlying process for events in the control sample.

Our control sample consists of generated $e^+e^- \to c\bar{c}$ Monte-Carlo events, where the signal and tag decays are chosen in the $D^{*+} \to [D^0 \to K_L^0\pi^+\pi^-]\pi^+$ and $D^{*-} \to [\bar{D}^0 \to K^+\pi^-]\pi^-$ channels, respectively. Since these events contain only one K_L^0, energy-momentum conservation can be used to calculate its four-momentum, which shows good positive correlation with the generated momentum. In this analysis, a reconstructed neutral KLM cluster candidate within a 10° cone in θ vs. ϕ plane of the calculated direction, and momentum within 25% of the calculated momentum, is considered as the signal K_L^0 candidate and the remaining ones are background.

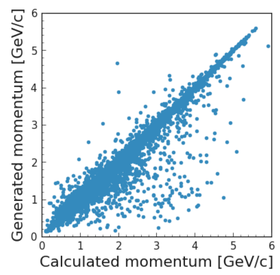


Figure 3: Correlation between the momentum of the calculated and generated K_L^0 candidates.

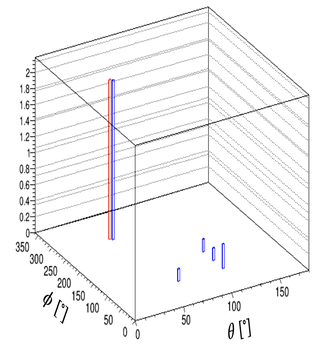


Figure 4: Reconstructed (calculated) momenta of K_L^0 candidates in blue (red) in θ vs. ϕ plane.

Signal and Background Probability Density Functions for Δm

The distribution of the discriminating variable $\Delta M = M_{D^{*+}} - M_{D^0}$ is obtained using the K_L^0 momentum as measured by the KLM detector, and fitted to a sum of signal and background probability density functions (PDF). The signal PDF has a peaking structure described by a double Gaussian, whereas the background PDF is a polynomial with an exponential damping term. This allows us to obtain the prior probabilities to study independent control variables.

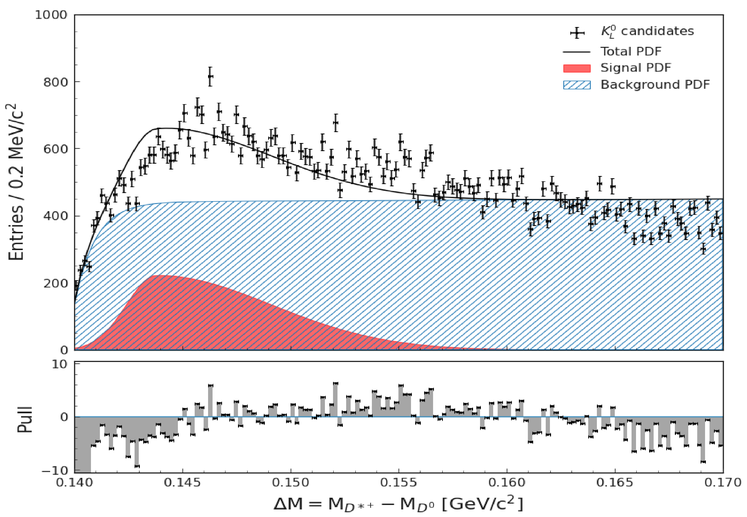


Figure 5: The ΔM distribution of reconstructed K_0^L candidates fitted to sum of signal and background PDFs.

Conventional vs $_sPlot$ techniques for K_L^0 identification

Each reconstructed K_L^0 is assigned a K-Long ID variable by particle identification tools, the distribution of which is shown in the left plot below for signal and background candidates based on the previously mentioned criteria.

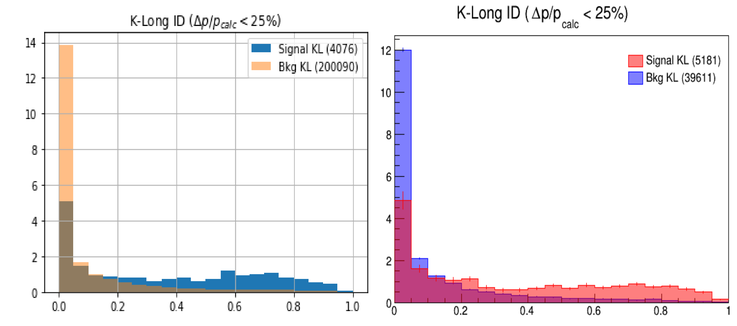


Figure 6: Conventional (left) based separation of K-Long ID distributions vs $_sPlot$ (right).

An alternative to this conventional technique is to unfold the total K-Long ID distribution using the signal and background prior probabilities obtained by fitting the ΔM distribution using the $_sPlot$ technique [5]. Given a well-defined knowledge of components of distributions of the discriminating variable, this technique allows unfolding the distribution of control variables into signal and background components, where in this analysis ΔM is the discriminating variable and K-Long ID is the control variable. This technique makes optimal use of uncertainties in a maximum likelihood based fit that provides the signal and background yields.

As evident from the above two distributions obtained using these two techniques, the $_sPlot$ yields considerably higher statistics available to study K_L^0 reconstruction efficiency. The difference between the signal and background candidates is observed more prominently for the candidates with K-Long ID > 0.2.

Summary and Outlook

Energy-momentum conservation in $e^+e^- \to c\bar{c}$ events in the chosen decay modes allows matching of reconstructed K_L^0 candidates with calculated four-momentum estimates, thereby defining signal and background categories. The $_sPlot$ based statistical technique is then used to study K_L^0 identification efficiency based on the distributions of K-Long ID variable from samples consisting of signal and background candidates. The salient feature of this new method is to study K_L^0 identification efficiency based on reconstructed level quantities without relying on generator level quantities. Although the method has been developed and validated using Monte-Carlo events, it eventually can be applied on real data using reconstructed level quantities only, as done for example at the *BABAR* experiment [6].

References

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